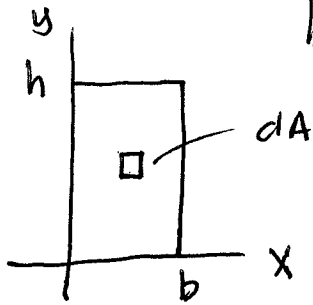


MoI and POI for Rectangle



MoI

$$I_x = \int_A y^2 dA$$

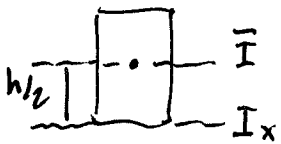
$$= \int_0^b \int_0^h y^2 dy dx$$

$$I_x = \frac{1}{3} b h^3$$

TI-89 $\int((y^2, y, 0, h), x, 0, b) \Rightarrow I_x = .333 b h^3$

To get $\bar{I}$:

$$I_x = \bar{I} + A d^2$$



$$\bar{I} = I_x - A d^2 = \frac{1}{3} b h^3 - (b h) (h/2)^2$$

$$= \left(\frac{1}{3} - \frac{1}{4} \right) b h^3 = \left(\frac{4}{12} - \frac{3}{12} \right) b h^3$$

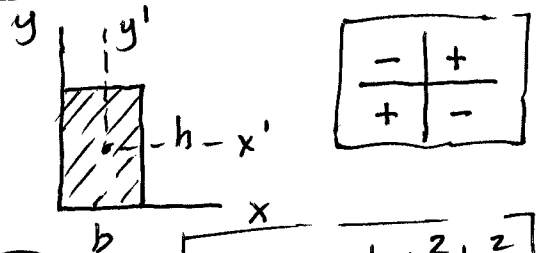
$$\bar{I} = \frac{1}{12} b h^3$$

POI

$$I_{xy} = \int_A x y dA$$

$$= \int_0^b \int_0^h x y dy dx$$

$$I_{xy} = \frac{1}{4} b^2 h^2$$



Touching x and y axes

$$I_{xy} = \frac{1}{4} b^2 h^2$$

$$= I_{ORIGIN}!$$

$$I_{xy} = \bar{I}_{xy} + A d_x d_y$$

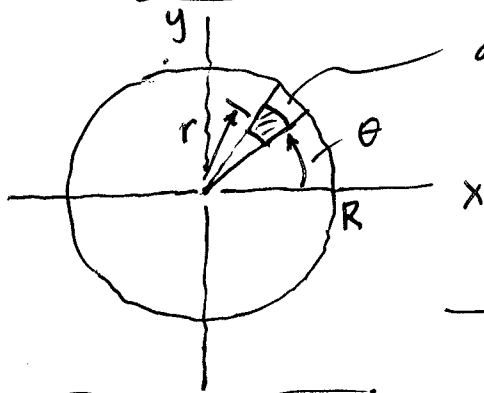
$$\bar{I}_{xy} = 0$$

About centroidal axes

Check: Is $I_{xy} = A d_x d_y$?

$$= (b h) \left(\frac{b}{2} \right) \left(\frac{h}{2} \right) = \frac{1}{4} b^2 h^2 \quad \text{yes!}$$

Integrate Area MOI for a Circle



$$dA = (r d\theta) dr$$

$$I_x = \int_A y^2 dA$$

$$x = r \cos \theta$$

$$[y = r \sin \theta]$$

$$I_x = \int_0^R \int_0^{2\pi} (r^2 \sin^2 \theta) \underbrace{r d\theta dr}_{dA}$$

Remember:

Set TI-89 mode to radians!

For calculator, let $r = K$
 $\theta = z$

$2 * \pi$

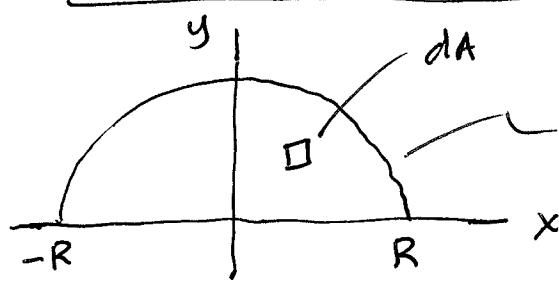
TI-89

$$\int(\int(K^3 * (\sin(z))^2, z, 0, 2\pi), K, 0, r)$$

Result: $I_x = .7854 R^4$

$$I_x = \frac{\pi}{4} R^4$$

Half - Circle MOI



$$x^2 + y^2 = R^2$$

$$y = (R^2 - x^2)^{.5}$$

$$I_x = \int_A y^2 dA$$

TI-89 $I_x = \int (\int (y^2, y, 0, (r^2 - x^2)^{.5}), x, -r, r)$

$$I_x = 0.3927 R^4 = I_{BASE}$$

$$= \frac{\pi}{8} R^4$$

$$I_y = \int_A x^2 dA$$

$$I_y = \int (\int (x^2, y, 0, (r^2 - x^2)^{.5}), x, -r, r)$$

$$I_y = 0.3927 R^4$$

$$= \frac{\pi}{8} R^4$$

To find $\bar{I}$, use PAT:



$$I_x = \bar{I} + A d^2$$

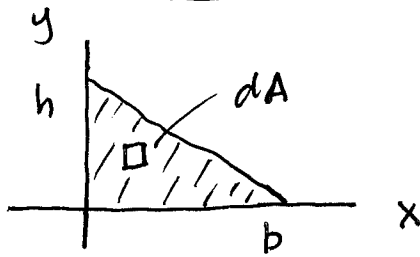
$$\bar{I} = I_x - A d^2$$

$$= \frac{\pi}{8} R^4 - \left(\frac{\pi R^2}{2} \right) \left(\frac{4R}{3\pi} \right)^2$$

$$= \left[\frac{\pi}{8} - \frac{\pi}{2} \left(\frac{16}{9\pi} \right) \right] R^4 = \left[\frac{\pi}{8} - \frac{8}{9\pi} \right] R^4$$

$$\bar{I} = 0.1098 R^4$$

MOI I_x for a Right Triangle



$$m = -h/b$$

$$y = (h - \frac{h}{b}x)$$

$$I_x = \int_A y^2 dA$$

$$I_x = \int_0^b \int_0^{(h - h/bx)} y^2 dy dx$$

ALPHA $\frac{h}{b}$

TI-89

$$\int(\int(y^2, y, 0, (h - (h/b)*x)), x, 0, b)$$

$$I_x = 0.08333 bh^3 = I_{BASE}$$

$$= \frac{1}{12} bh^3$$

To find $\bar{I}$ about centroidal axis:

Use PAT:

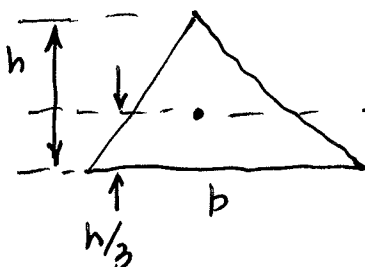
$$I_x = \bar{I} + Ad^2$$

$$\frac{1}{12}bh^3 = \bar{I} + (\frac{1}{2}bh)(\frac{h}{3})^2$$

$$\bar{I} = bh^3 \left(\frac{1}{12} - \frac{1}{18} \right)$$

$$= \left(\frac{3}{36} - \frac{2}{36} \right) = \frac{1}{36}$$

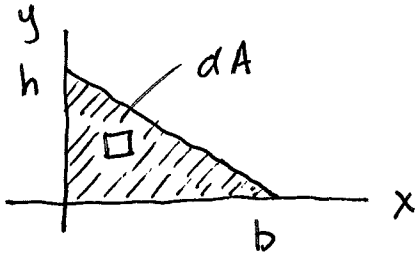
$$\bar{I} = \frac{1}{36} bh^3$$



$$\bar{I} = \frac{1}{36} bh^3$$

$$I_{Base} = \frac{1}{12} bh^3$$

POI I_{xy} for a Right Triangle



$$m = -h/b$$

$$y = h - (h/b)x$$

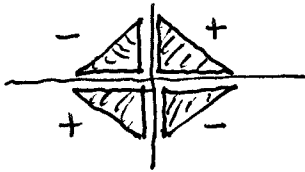
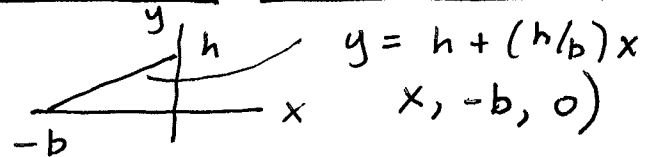
$$I_{xy} = \int_A xy \, dA$$

$$I_{xy} = \int_0^b \int_0^{h-(h/b)x} xy \, dy \, dx$$

TI-89 $\int(\int(x*y, y, 0, (h-(h/b)*x)), x, 0, b)$

$$I_{xy} = 0.04167 b^2 h^2$$

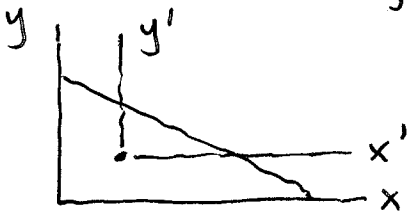
Different orientations
(quadrants)



$$I_{xy} = -0.04167 b^2 h^2$$

★ If right triangle is at origin!

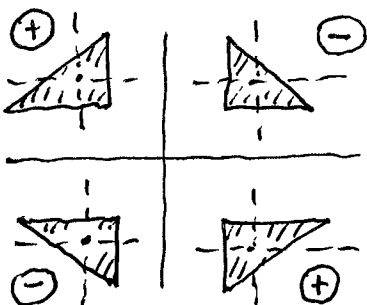
To find $\bar{I}_{xy}$ about centroidal axes:



Use PAT: $I_{xy} = \bar{I}_{xy} + A d_x d_y$

$$.04167 b^2 h^2 = \bar{I}_{xy} + (\frac{1}{2}bh)(\frac{b}{3})(\frac{h}{3})$$

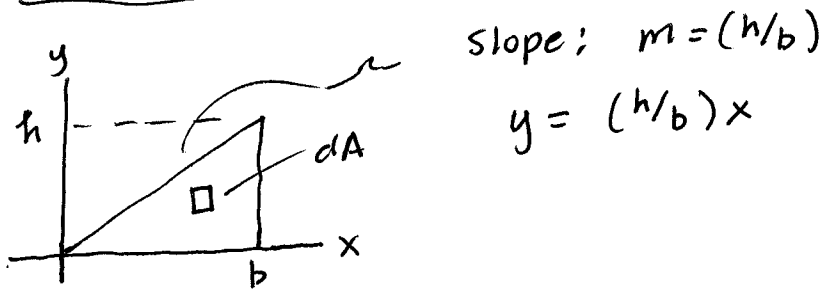
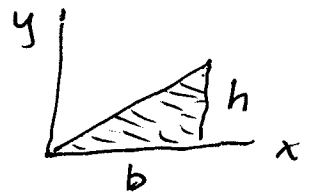
$$\bar{I}_{xy} = [.04167 - \frac{1}{18}] b^2 h^2$$



$$\bar{I}_{xy} = -0.013886 b^2 h^2$$

$$= -\frac{1}{72} b^2 h^2$$

Hibbeler's I_x and I_y (MOI)
for a Triangle at this Orientation



$$I_x = \int_A y^2 dA$$

$$I_y = \int_A x^2 dA$$

$$I_x = \int_0^b \int_0^{(h/b)x} y^2 dy dx$$

$$I_x = \int \left(\int (y^2, y, 0, (h/b) * x), x, 0, b \right)$$

$$I_x = 0.0833 bh^3$$

$$= \frac{1}{12} bh^3$$

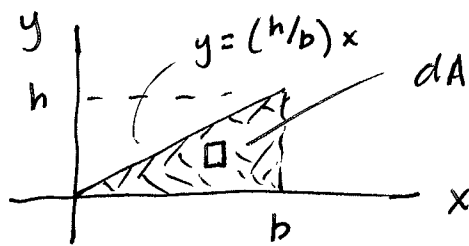
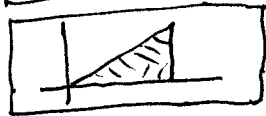
$$I_y = \int \left(\int (x^2, y, 0, (h/b) * x), x, 0, b \right)$$

$$I_y = 0.25 hb^3$$

$$= \frac{1}{4} hb^3$$

at this
orientation

Hibbeler's
POI for a
Right Triangle



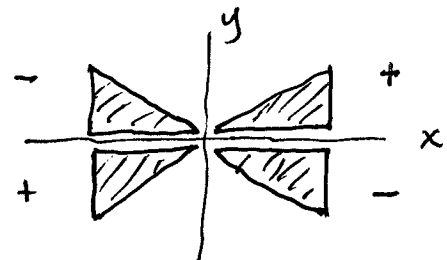
$$I_{xy} = \int_A xy \, dA$$

$$I_{xy} = \int_0^b \int_0^{(h/b)x} xy \, dy \, dx$$

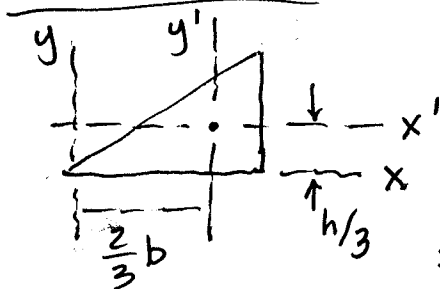
TI-89 $\int(\int(x*y, y, 0, (h/b)*x), x, 0, b)$

$$I_{xy} = 0.125 b^2 h^2$$

$$= \frac{1}{8} b^2 h^2$$



for a right triangle at
this orientation } that touches the
x-y axes!!



To find $\bar{I}_{xy}$:

Use PAT : $I_{xy} = \bar{I}_{xy} + A dx dy$

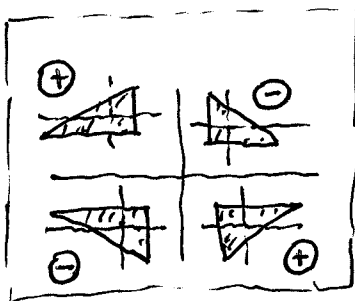
$$\bar{I}_{xy} = I_{xy} - A dx dy$$

$$= \frac{1}{8} b^2 h^2 - \left(\frac{1}{2} bh \right) \left(\frac{2}{3} b \right) \left(\frac{h}{3} \right)$$

$$= \left[\frac{1}{8} - \frac{1}{9} \right] b^2 h^2$$

$$= \left(\frac{9}{72} - \frac{8}{72} \right)$$

$$= \left(\frac{1}{72} \right)$$

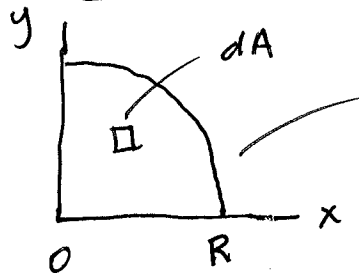


$$\bar{I}_{xy} = + \frac{1}{72} b^2 h^2$$

at this
orientation

Centroidal
 $\bar{I}_{xy}$ for Right Triangle

Quarter Circle MOI



$$x^2 + y^2 = R^2$$

$$y = (R^2 - x^2)^{.5}$$

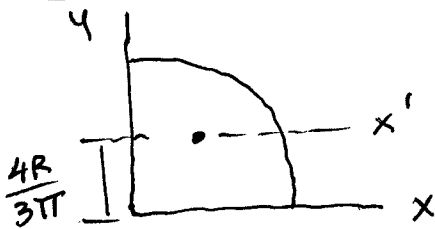
$$I_x = \int_A y^2 dA$$

$$I_x = \int_0^R \int_0^{(R^2 - x^2)^{.5}} y^2 dy dx$$

TI-89 $I_x = \int (\int (y^2, y, 0, (R^2 - x^2)^{.5}), x, 0, R)$

$$I_x = 0.19635 R^4 = I_{BASE}$$

$$= \frac{\pi}{16} R^4$$



To find $\bar{I}$ about centroidal axis:

Use PAT: $I_x = \bar{I} + A d^2$

$$\bar{I} = I_x - A d^2 = \frac{\pi}{16} R^4 - \left(\frac{\pi R^2}{4} \right) \left(\frac{4R}{3\pi} \right)^2$$

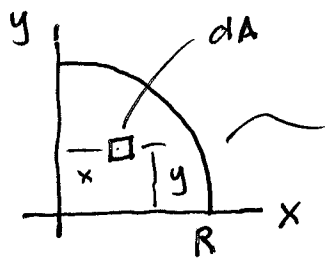
$$\bar{I} = \left[\frac{\pi}{16} - \frac{\pi}{4} \left(\frac{16}{9\pi} \right) \right] R^4$$

$$\bar{I} = \left[\frac{\pi}{16} - \frac{4}{9\pi} \right] R^4$$

$$\bar{I} = .05488 R^4$$

About centroidal axis

Quarter Circle POI



Circle: $x^2 + y^2 = R^2$
 $y = (R^2 - x^2)^{.5}$

$$I_{xy} = \int_A \tilde{x} \tilde{y} dA$$

$$dA = dx \cdot dy$$

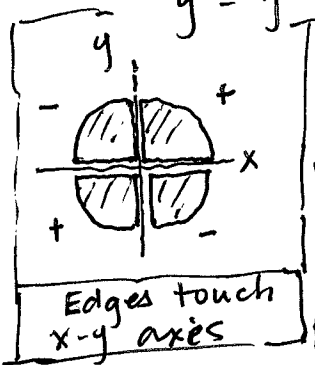
$$\tilde{x} = x$$

$$\tilde{y} = y$$

$$I_{xy} = \int_0^R \int_0^{(R^2 - x^2)^{.5}} xy \, dy \, dx$$

[ALPHA] + r

TI-89: $I_{xy} = \int ((x \times y, y, 0, (r^2 - x^2)^{.5}), x, 0, r)$



$$I_{xy} = 0.125 R^4$$

$$= + \frac{1}{8} R^4$$

This about the x-y axes along its edges.

To find $\bar{I}_{xy}$ about its centroidal axes:

Use PAT:

$$I_{xy} = \bar{I}_{xy} + A dx dy$$

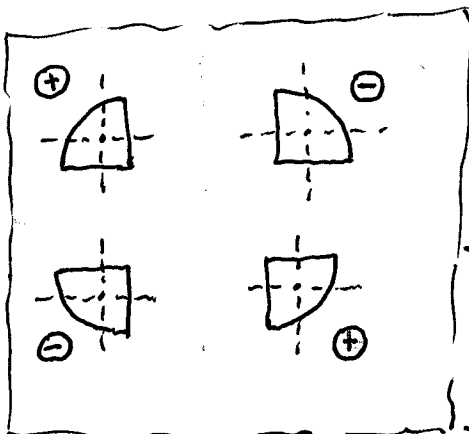
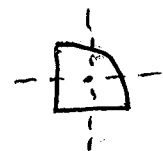
$$\bar{I}_{xy} = I_{xy} - A dx dy$$

$$= \frac{1}{8} R^4 - \left(\frac{\pi}{4} R^2 \right) \left(\frac{4R}{3\pi} \right) \left(\frac{4R}{3\pi} \right)$$

$$= \frac{1}{8} R^4 - \frac{\pi}{4} R^2 \frac{16 R^2}{9\pi}$$

$$= \left[\frac{1}{8} - \frac{4}{9\pi} \right] R^4$$

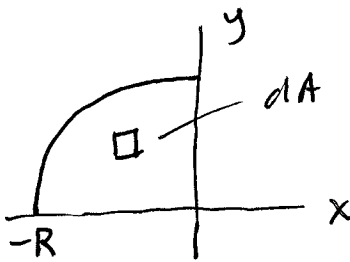
$$\bar{I}_{xy} = -0.01647 R^4$$



$\bar{I}_{xy}$ about centroidal axes!

Quarter Circle POI

Another orientation



$$y = (R^2 - x^2)^{.5}$$

$$I_{xy} = \int_A xy \, dA$$

$$I_{xy} = \int_{-R}^0 \int_0^{(R^2 - x^2)^{.5}} xy \, dy \, dx$$

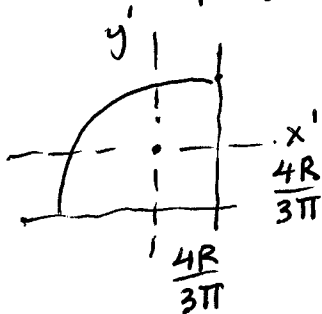
limits on
x change

$$I_{xy} = \int \left(\int (x \cdot y, y, 0, (R^2 - x^2)^{.5}), x, -R, 0 \right)$$

$$I_{xy} = -\frac{1}{8} R^4$$

NEG! at this orientation

Find $\bar{I}_{xy}$ about centroidal axes:



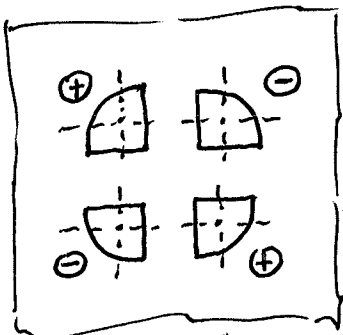
$$I_{xy} = \bar{I}_{xy} + A \, dxdy$$

$$\bar{I}_{xy} = I_{xy} - A \, dxdy$$

$$= \left(-\frac{1}{8} R^4 \right) - \left(\frac{\pi}{4} R^2 \right) \left(-\frac{4R}{3\pi} \right) \left(+\frac{4R}{3\pi} \right)$$

$$= \left[-\frac{1}{8} + \cancel{\frac{\pi}{4}} R^4 \frac{16R^2}{9\pi} \right]$$

$$= \left[-\frac{1}{8} + \frac{4}{9\pi} \right] R^4$$



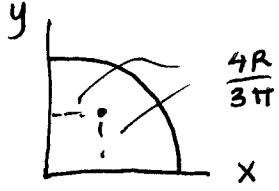
$$\bar{I}_{xy} = +0.01647 R^4$$

at this orientation

About centroidal axes

Centroidal I_{xy} Values (Quarter Circle)

Given:


$$I_{xy} = +\frac{1}{8} R^4$$

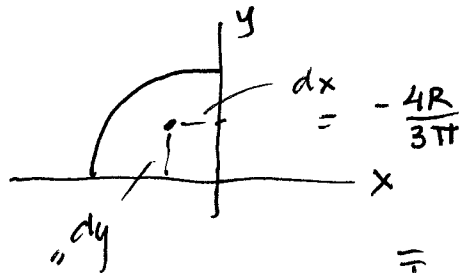
$$I_{xy} = \bar{I}_{xy} + A dx dy$$

$$\begin{aligned}\bar{I}_{xy} &= I_{xy} - A dx dy \\ &= \frac{1}{8} R^4 - \left(\frac{\pi R^2}{4} \right) \left(\frac{4R}{3\pi} \right) \left(\frac{4R}{3\pi} \right)\end{aligned}$$

$$\bar{I}_{xy} = \left[\frac{1}{8} - \frac{4}{9\pi} \right] R^4$$

$$\boxed{\bar{I}_{xy} = -0.01647 R^4}$$

Given



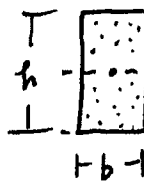
$$\boxed{I_{xy} = -\frac{1}{8} R^4}$$

$$\begin{aligned}\bar{I}_{xy} &= I_{xy} - A dx dy \\ &= -\frac{1}{8} R^4 - \left(\frac{\pi R^2}{4} \right) \left(-\frac{4R}{3\pi} \right) \left(+\frac{4R}{3\pi} \right) \\ &= \left[-\frac{1}{8} + \frac{4}{9\pi} \right] R^4\end{aligned}$$

$$\boxed{\bar{I}_{xy} = +0.01647 R^4}$$

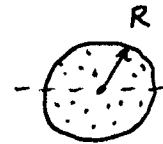
etc.

(MOI) Moments of Inertia of Common Shapes (and POI Values)

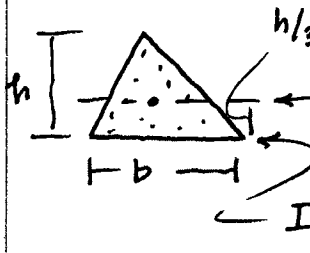


$$\bar{I} = \frac{1}{12} b h^3$$

$$I_{base} = I_b = \frac{1}{3} b h^3$$

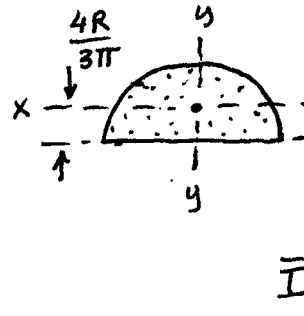


$$\bar{I} = \frac{1}{4} \pi R^4$$



$$\bar{I} = \frac{1}{36} b h^3$$

$$I_{base} = \frac{1}{12} b h^3$$



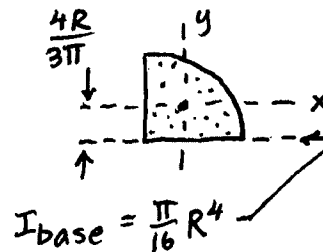
$$I_{base} = \frac{\pi}{8} R^4$$

$$\bar{I}_{xx} = .1098 R^4$$

$$\bar{I}_{yy} = \frac{1}{8} \pi R^4$$

Parallel Axis Theorem (PAT):

$$I = \bar{I} + A d^2$$



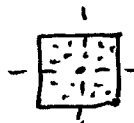
$$I_{base} = \frac{\pi}{16} R^4$$

$$\bar{I}_{xx} = \bar{I}_{yy} = 0.05488 R^4$$

(POI) Product of Inertia for Common Areas (Centroidal Values)

$\bar{I}_{xy} = 0$ for areas symmetric about either the $\bar{x}$ or $\bar{y}$ axes:

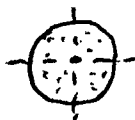
PAT: $I = \bar{I}_{xy} + A d_x d_y$



$$\bar{I}_{xy} = 0$$



$$\bar{I}_{xy} = 0$$



$$\bar{I}_{xy} = 0$$



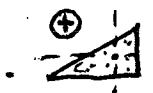
$$\bar{I}_{xy} = 0$$

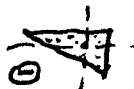
Note: I_{base} not defined for I_{xy} , bc I_{xy} is relative to a pair of orthog axes

Non-Symmetric Areas (Centroidal $\bar{I}_{xy}$ Values)

Right Triangles

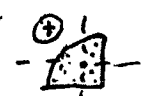
$n =$ Quadrant




$$\bar{I}_{xy} = (-1)^n \frac{1}{72} b^2 h^2$$

$n = \text{quadrant}$







$$\bar{I}_{xy} = (-1)^n .01647 R^4$$

$x-y$
 $x'-y'$
 $u-v$